

Dietary flavonoid intake and breast cancer survival among women on Long Island.

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BACKGROUND: Laboratory research and a growing number of epidemiologic studies have provided evidence for a reduced risk of breast cancer associated with dietary intake of certain classes of flavonoids. However, the effects of flavonoids on survival are not known. In a population-based cohort of breast cancer patients, we investigated whether dietary flavonoid intake before diagnosis is associated with subsequent survival. METHODS: Women ages 25 to 98 years who were newly diagnosed with a first primary invasive breast cancer between August 1, 1996, and July 31, 1997, and participated in a population-based, case-control study (n=1,210) were followed for vital status through December 31, 2002. At the case-control interview conducted shortly after diagnosis, respondents completed a FFQ that assessed dietary intake in the previous 12 months. All-cause mortality (n=173 deaths) and breast cancer-specific mortality (n=113 deaths) were determined through the National Death Index. RESULTS: Reduced hazard ratios [age- and energy-adjusted hazard ratio (95% confidence interval)] for all-cause mortality were observed among premenopausal and postmenopausal women for the highest quintile of intake, compared with the lowest, for flavones [0.63 (0.41-0.96)], isoflavones [0.52 (0.33-0.82)], and anthocyanidins [0.64 (0.42-0.98)]. No significant trends in risk were observed. Results were similar for breast cancer-specific mortality only. CONCLUSION: Mortality may be reduced in association with high levels of dietary flavones and isoflavones among postmenopausal U.S. breast cancer patients. Larger studies are needed to confirm our findings.